

# **HL Mathematics: Analysis and Approaches**

## *Epidemic Theory*

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# **1. Introduction**

## **1.1 Rationale**

For the past two years, I have turned on the news only to see the same familiar headline: “Breaking News on the COVID-19 Pandemic. I became increasingly aware that, as the months passed, there would always be a new country that would be broadcasted on the news with surging cases of the virus, only for them to fade out and be replaced by a new hotspot. While the case numbers and locations were seemingly random globally, it was evident that there was a repetition in the pattern of infection in these different countries.

When it came the time to choosing a topic for this exploration, I was naturally drawn to this topic and stumbled upon one of epidemic theory’s most prominent subtopic: SIR Epidemic models. Dating back to as early as 1916 with the works of Ronald Ross, SIR Epidemic models aim to quantify and generalize the spread of a disease through a population through the use of differential equations. Nonetheless, the original model continues to serve as an accurate model to approximate the spread of diseases.

## **1.2 Aim**

This mathematics exploration aims to delineate the method by which an SIR model can be generated, with respect to three main time-dependent variables ( $S(t)$ ,  $I(t)$ ,  $R(t)$ ), as well as derive the base conditions and significant values concerning disease spread. The model’s efficacy will subsequently be analyzed by fitting the obtained differential equations with real world data from the 2020-21 COVID-19 virus spread in Singapore. By plotting and comparing the generated model to the actual progression of infections, the accuracy and therefore applicability of SIR epidemic models can be determined.

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## 2. Background and Overview

### 2.1 Epidemic Theory

**Epidemic theory** is a branch of mathematics concerning the *creation of statistical models to visually quantify the incidence and spread of a communicable infectious disease*<sup>[1]</sup>, using data such as environmental conditions, host resistance, etc. to create epidemic trends modelling as many relevant real-world conditions as possible.

### 2.2 SIR Epidemic Modelling

SIR (Susceptible-Infected-Recovered) Epidemic models detail the progression of a given disease by dividing a region's population ( $N$ ) into three time-dependent *compartments*:

- Susceptible,  $s(t)$ : Members of the population who are susceptible to the disease
- Infected,  $i(t)$ : Members of the population who have been infected and are infectious
- Recovered/Removed,  $r(t)$ : Members of the population who have recovered or passed away from the disease and are no longer part of any equations.

The SIR model also makes use of two independent variables, *infection rate* ( $\lambda$ ) and *recovery rate* ( $\beta$ ), which help determine the population's progresses from  $S(t)$  to  $I(t)$  and subsequently to  $R(t)$ .

This exploration aims to derive the three compartment's differential equations, analyze them to mathematically quantify key features of an epidemic, and graph them using Euler's Method. Finally, the equations will be applied to a historical epidemic, to check for accuracy of the predicted trends with the real-world ones.

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<sup>[1]</sup> *epidemic theory*. (n.d.). Oxford Reference. Retrieved December 13, 2021, from <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095754542?rskey=yBZ9Yn&result=9>

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### 3. Mathematical Derivation of the model

#### 3.1 Conditions presupposing the model

1. The sum of the people in the susceptible, infected, and recovered/removed categories (population) should remain the same, changes due to birth/death rate and net migration are neglected.
2. The rate of infection should be proportional to the number of contacts between infected individuals ( $I(t)$ ) and susceptible individuals ( $S(t)$ ).
3. Rate of transition between infected ( $I(t)$ ) and recovered/removed ( $R(t)$ ) categories should remain constant.

#### 3.2 Variables involved

SIR Epidemic models need to be set such that they can be applied for countries of different sizes, thus the values of susceptible, infected, and recovered/removed need to be adjusted to account for population differences. To obtain the proportions, each value is divided by  $N$ , the population size.

$$S(t) = \frac{s(t)}{N}, \quad I(t) = \frac{i(t)}{N}, \quad R(t) = \frac{r(t)}{N}$$

Thus, we get three new variables:

**S(t):** Population density part of the *susceptible* category at a given point in time

**I(t):** Population density part of the *infected* category at a given point in time

**R(t):** Population density part of the *removed* at a given point in time

From the first assumption of a constant population, we obtain the expression:

$$\begin{aligned} s + i + r &= N \\ \Rightarrow \frac{s}{N} + \frac{i}{N} + \frac{r}{N} &= \frac{N}{N} \end{aligned}$$

---


$$\therefore S + I + R = 1$$

When  $t = 0$ , All individuals are either Susceptible or Infected <sup>[2]</sup>

$$\therefore S(0) + I(0) = 1, S_0 + I_0 = 1 \text{ }^{[3]}$$

### 3.3 Modeling Rate of Change using differential equations

According to the second condition, the rate of infections should be proportional to the number of contacts between infected and susceptible individuals.

$$\Rightarrow \text{New Infections} \propto \text{Contacts}$$

$$\Rightarrow \text{New Infections} = k (S \times I), \text{ for some } k \in R$$

$$\therefore \text{New Infections} = \lambda (SI), \text{ where } \lambda \text{ is the infection rate}$$

According to the third condition, the infectives transition to the recovered/removed category at a constant rate throughout the epidemic.

$$\Rightarrow \text{Recoveries/Removals} \propto \text{Infections}$$

$$\Rightarrow R = m(I) \text{ (for some } m \in R)$$

$$\therefore R = \beta(I), \text{ where } \beta \text{ is the recovery/removal rate of the particular disease}$$

People who move into infected category move out of the susceptible category.

$$\Rightarrow S_{t+1} = S_t - \lambda(SI)$$

$$S_{t+1} - S_t = -\lambda(SI)$$

$$\therefore \frac{dS}{dt} = -\lambda(SI) \text{ ----- [1]}$$

People who move into the recovered/removed category move out of the infected one

$$\therefore R_{t+1} = R_t + \beta(I)$$

$$R_{t+1} - R_t = \beta(I)$$

$$\therefore \frac{dR}{dt} = \beta(I) \text{ ----- [2]}$$

---

<sup>[2]</sup>  $R(0) = 0$ ,  $\therefore$  the value is not considered in any calculations

<sup>[3]</sup>  $S_0, I_0$  are important variables that will be considered later in this exploration

---

The infected category comprises of those who move out of the susceptible category minus those who move into the recovered/removed category,

$$\therefore I_{t+1} = I_t - \beta(I) + \lambda(SI)$$

$$I_{t+1} - I_t = \lambda(SI) - \beta(I)$$

$$\therefore \frac{dI}{dt} = \lambda(SI) - \beta(I) \text{ ----- [3]}$$

$$S + I + R = 1$$

$$\therefore \frac{d}{dt}(S + I + R) = \frac{d}{dt}(1)$$

$$\frac{d}{dt}(S + I + R) = 0$$

$$\therefore \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0$$

$$= [-\lambda(SI)] + [\lambda(SI) - \beta(I)] + [\beta(I)] = 0$$

### 3.4 Threshold Values

By this point in my exploration, I had obtained the base equations to plot and compare the compartmental trends. However, reflecting on my journey thus far, I wanted to delve deeper into the relationship between the equations and their use cases in theoretically predicting key features of an epidemic spread. Thus, I was led to explore the threshold values derived from the compartmental equations.

#### 3.4.1 Base Reproductive Rate ( $R_0$ )

The number of susceptible people will always decrease throughout the epidemic period,  $\therefore S \leq S_0$

Putting in  $S_0$  into equation [1]:

$$\frac{dI}{dt} < \lambda(S_0 I) - \beta(I)$$

$$= \frac{dI}{dt} < I(\lambda S_0 - \beta)$$



---

For the disease to spread, rate of change must be positive,  $\therefore \frac{dI}{dt} > 0$

$$\therefore I(\lambda S_0 - \beta) > 0$$

$$\lambda S_0 - \beta > 0$$

$$\therefore S_0 > \frac{\beta}{\lambda}$$

$$S_0 \left(\frac{\lambda}{\beta}\right) > \left(\frac{\beta}{\lambda}\right) \cdot \left(\frac{\lambda}{\beta}\right)$$

$$\therefore S_0 \frac{\lambda}{\beta} > 1 \text{ (This value is known as the Base Reproductive Rate, } R_0\text{)}$$

$$S_0 \frac{\lambda}{\beta} = R_0 \text{ ----- [4]}$$

$R_0$  is thus arguable the most important constant required under SIR Epidemic Models. The value lets us know whether a particular will spread to form an epidemic or not and can be used to describe several other features of an epidemic, as shown in the subsequent threshold value derivations.

### 3.4.2 Maximum Infection Density ( $\frac{I_{max}}{N}$ )

Using the original differential equation, the maximum infected density (maximum proportion of population infected at a point during the epidemic) can be found in terms of the base reproductive rate  $R_0$ .

Dividing equations (1) by (3):

$$\frac{\frac{dS}{dt}}{\frac{dI}{dt}} = \frac{-\lambda(SI)}{\lambda(SI) - \beta(I)}$$

$$\frac{dS}{dI} = \frac{-\lambda(SI)}{\lambda(SI) - \beta(I)}$$

$$\frac{dS}{dI} = \frac{-\lambda S}{\lambda S - \beta}$$

Given that this equation is a Separable Differential Equation,

$$\Rightarrow \int \frac{\lambda S - \beta}{\lambda S} dS = -\int dI$$

$$\int 1 - \int \frac{\beta}{\lambda S} dS = - \int 1 dI$$

$$\int 1 dS - \frac{\beta}{\lambda} \int \frac{1}{S} dS = - \int 1 dI$$

$$\therefore S - \beta \left( \frac{\ln(S)}{\lambda} \right) = -I + C \text{ (For some arbitrary constant C) ----- [5]}$$

$$C = S + I - \frac{\beta}{\lambda} \ln(S)$$

$$\text{When } t = 0, S = S_0 \text{ \& } I = I_0$$

$$\therefore C = S_0 + I_0 - \frac{\beta}{\lambda} \ln(S_0)$$

Thus, equation [5] can be expressed as:

$$S + I - \frac{\beta}{\lambda} \ln(S) = S_0 + I_0 - \frac{\beta}{\lambda} \ln(S_0) \text{ ----- [6]}$$

$$\text{At } I_{max}, \frac{dI}{dt} = 0. \text{ Therefore, } \lambda(SI) - \beta(I) = 0$$

$$\lambda S - \beta = 0$$

$$\therefore S = \frac{\beta}{\lambda} \text{ ----- [7]}$$

Substituting [7] into [6] and  $I = I_{max}$ ,

$$I_{max} + \frac{\beta}{\lambda} - \frac{\beta}{\lambda} \ln\left(\frac{\beta}{\lambda}\right) = S_0 + I_0 - \frac{\beta}{\lambda} \ln(S_0)$$

$$I_{max} = S_0 + I_0 - \frac{\beta}{\lambda} \ln(S_0) - \frac{\beta}{\lambda} + \frac{\beta}{\lambda} \ln\left(\frac{\beta}{\lambda}\right)$$

$$I_{max} = N - \frac{\beta}{\lambda} \left( -\ln\left(\frac{\beta}{\lambda}\right) + \ln(S_0) + 1 \right)$$

Assuming  $I_0 = 0, N = S_0$

$$\therefore \frac{\beta}{\lambda} \left( \frac{1}{N} \right) = \frac{\beta}{\lambda} \left( \frac{1}{S_0} \right) = \frac{1}{R_0}$$

$$I_{max} = N - \frac{\beta}{\lambda} \left( \ln(S_0 \left( \frac{\beta}{\lambda} \right)) + 1 \right)$$

$$\frac{I_{max}}{N} = 1 - \frac{\beta}{\lambda} \left( \frac{1}{N} \right) (\ln(R_0) + 1)$$

$$\therefore \frac{I_{max}}{N} = 1 - \frac{1}{R_0} (\ln(R_0) + 1) \text{ ----- [8]}$$

Thus, the maximum infection density has been expressed in terms of  $R_0$ . The presence of  $\ln(R_0)$  indicates the non-linear nature of this value- it scales logarithmically with a disease of higher  $R_0$  value. Aside from the maximum infection density, calculating the end of an epidemic is also important.

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### 3.4.3 Predicting Epidemic situation at $t = \infty$

It has traditionally been assumed that when  $S(t) = 0$ , an epidemic has finished. Using SIR models, this idea that an epidemic ends at some time  $t = \infty$  can be disproven. First, an expression for  $S(t)$  must be obtained.

Using the original differential equations and dividing equation (1) by (2),

$$\frac{\frac{dS}{dt}}{\frac{dR}{dt}} = \frac{-\lambda(SI)}{\beta(I)}, \therefore \frac{dS}{dR} = \frac{-\lambda(SI)}{\beta(I)}$$

$$\frac{dS}{dR} = \frac{-\lambda S}{\beta}$$

$$\therefore \int \frac{1}{S} dS = \int \frac{-\beta}{\lambda} dR$$

$$\ln|S| = \frac{-\beta}{\lambda}(R) + C \text{ (For some arbitrary constant } C) \text{ ----- [9]}$$

When  $t = 0$ ,  $R(0) = 0$  &  $S = S_0$

$$\therefore \ln(S_0) = C$$

Substituting this value of  $C$  into equation [9]:

$$\ln(S) = \frac{-\beta}{\lambda}(R) + \ln(S_0)$$

$$\ln(S) - \ln(S_0) = \frac{-\beta}{\lambda}(R)$$

$$\ln\left(\frac{S}{S_0}\right) = \frac{-\beta R}{\lambda}$$

$$\frac{S}{S_0} = e^{\frac{-\beta R}{\lambda}}$$

$$\therefore S(t) = S_0 e^{\frac{-\beta R(t)}{\lambda}} \text{ ----- [10]}$$

Now that we have obtained an expression for  $S(t)$ , we can talk about  $S_\infty$

Given that  $0 \leq R(t) \leq N$ ,  $e^{\frac{-\beta R(t)}{\lambda}} < 1$ .

$$\therefore S(t) \geq S_0 e^{\frac{-\beta N}{\lambda}} > 0 \text{ ----- [11]}$$

At  $t = \infty$ ,  $S(t) = S_\infty$

Substituting this into expression [11];

$$S_{\infty} \geq S_0 e^{-S_0} > 0 \text{ ----- [12]}$$

$$\therefore S_{\infty} \neq 0$$

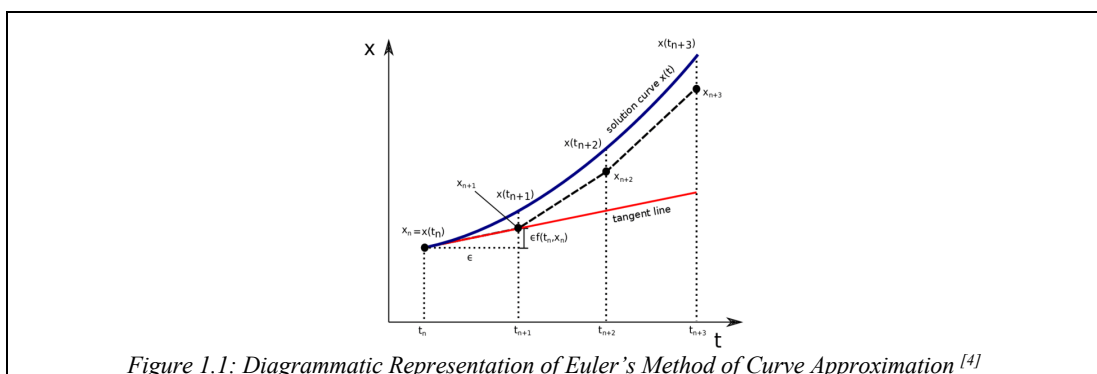
Thus, the total susceptible population is never truly 0. Rather, it is when the number of infected individuals tends to zero than a region begins to reach an *endemic*, or the terminal stages of an epidemic.

Now that the key features of an epidemic have been analyzed, the three compartments need to be visually represented. There is no way of representing  $S(t)$ ,  $I(t)$ , and  $R(t)$  mathematically, thus necessitating the use of Euler's Method to plot the trends.

## 4. Euler's Method of Trend Plotting

### 4.1 Euler's Method Overview

Euler's Method is a numerical procedure used to numerically represent differential equations which cannot be solved analytically. A curve can be plotted by finding the value of the tangent at periodic intervals/steps ( $\Delta x$ ), and thus by mapping out the tangents a curve can be approximated, as shown on *Figure 1.1* below. Euler's first order method implies that that the error in approximation of the differential equation per step is proportional to the square of the step size.



<sup>4</sup> ResearchGate. (2019, July). *Euler's Method* [Graph]. ResearchGate. [https://www.researchgate.net/figure/Eulers-method-The-dashed-line-shows-the-solution-computed-by-successive-iterations-of\\_fig1\\_333678760](https://www.researchgate.net/figure/Eulers-method-The-dashed-line-shows-the-solution-computed-by-successive-iterations-of_fig1_333678760)

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To approximate a differential equation with a given step  $\Delta x$  using Euler's method:

$$y(x + \Delta x) = y(x) + \Delta x \left( \frac{dy}{dx} \right)$$

#### 4.2 Using Euler's Method to approximate the Differential Equations

Given that daily changes are being considered,  $\Delta x = \Delta t = 1$

For the susceptible column,

$$S_{t+\Delta t} = S_t + \Delta t \left( \frac{dS}{dt} \right)$$

$$\therefore S_{t+1} = S_t - 1 (\lambda(SI))$$

For the infected column,

$$I_{t+\Delta t} = I_t + \Delta t \left( \frac{dI}{dt} \right)$$

$$\therefore I_{t+1} = I_t + 1 (\lambda(SI) - \beta(I))$$

For the recovered column,

$$R_{t+\Delta t} = R_t + \Delta t \left( \frac{dR}{dt} \right)$$

$$\therefore R_{t+1} = R_t + 1 (\beta(I))$$

The three equations generate a model, not an analytical solution. However, using some standard formulae, these trends can be visually approximated.

#### 4.3 Visually Approximating Trends to Mathematical Functions

Euler's method is a numerical approximation as opposed to analytical solution with a definite function. Having studied exponential graphs in school and sigmoidal graphs in the context of this exploration, I realized that the two graphs follow similar shapes to the SIR compartments. Using the Desmos online software, I visually approximated the three trends. Considering a sample 160-day epidemic with  $\lambda = 0.2$  and  $\beta = 0.1$ :

$$S(t) = -\frac{1}{1+e^{-\left(\frac{x-80}{12}\right)}} + 1 \text{ (Negative Sigmoidal Function of form } -\frac{1}{1+e^{-x}}\text{)}$$

$$I(t) = e^{-\left(\frac{x+31.318}{110}-1\right)^2} - 0.88 \text{ (Exponential Function of form } e^{-x^2}\text{)}$$

$$R(t) = \frac{1}{1+e^{-\frac{x-80}{12}}} \text{ (Sigmoidal Function of form } \frac{1}{1+e^{-x}}\text{)}$$

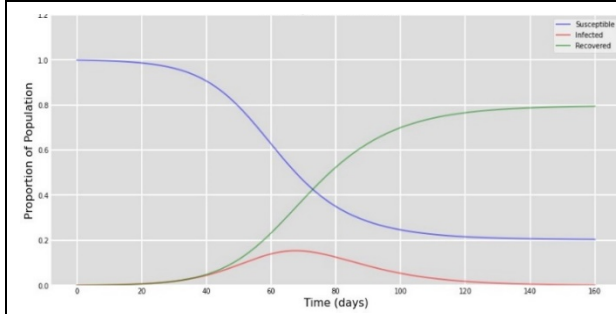


Fig. 1.1: Generic shapes of the different compartmental trends part of the SIR Epidemic Model.

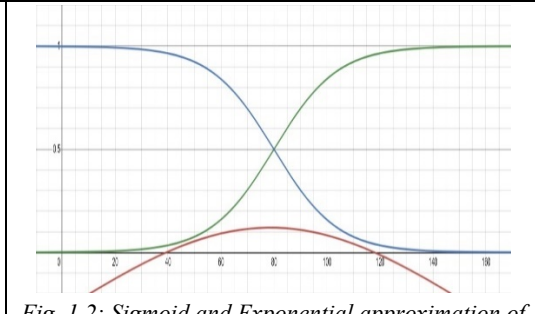


Fig. 1.2: Sigmoid and Exponential approximation of the different compartmental trends part of the SIR Epidemic Model. <sup>[5]</sup>

Thus, the formula approximations explain the sigmoidal and exponential trends of the susceptible, infected, and recovered trends over the course of an epidemic.

Now that we have mathematically broken down and approximated each trend, we can analyze the accuracy of SIR models by fitting them with real-world data from a historical epidemic.

## 5. Case Study: COVID-19 Pandemic in Singapore, 2020-2021

### 5.1 Overview

Originating in the Chinese Province Wuhan in December 2019, the COVID-19 pandemic has infected more than 264 million people globally and claimed 5.23 million lives. Singapore was one such affected country- despite taking several precautionary measures to halt virus transmission, they nonetheless fell victim to the spread.

For the purposes of this analysis, the epidemic will be considered from the 23<sup>rd</sup> of January 2020 to the 11<sup>th</sup> of July 2021. These 536 days have been chosen in particular

<sup>[5]</sup> Mathematical trends plotted using Desmos Graphing Software

as, after the 11<sup>th</sup> of July, Singapore saw a second surge in cases, which would color the comparison and analysis of this exploration.

## 5.2 Application of Euler's Method to plot the SIR model with 2020-21 Singapore COVID-19 Pandemic Data

Recovery time for COVID-19: 14 days

Median time for death after first infection: 18.5 days

Proportion of infected who pass away: 0.28%

$$\therefore \beta = \frac{1}{(14(1-0.0028)+18.5(0.0028))}; \beta = 0.071$$

Number of people infected on first day: 1

$$\therefore i_0 = 1$$

Population of Singapore in 2020: 5.686 million

$$\therefore N = 5.686 \times 10^6$$

$$I_0 = \frac{i_0}{N} = \frac{1}{5686000} = 1.76 \times 10^{-7}$$

$$\therefore S_0 = 1 - 1.76 \times 10^{-7}; S_0 = 0.9999998241$$

Base Reproductive Rate of COVID-19 in Singapore: 1.5 <sup>[6]</sup>

$\therefore$  Using the expression for  $R_0$  (equation [4]):

$$\begin{aligned} \lambda &= \frac{1.5\beta}{S_0} \\ &= \frac{1.5(0.071)}{0.9999998241} = 0.107 \end{aligned}$$

Using the values and the original expression, the three trends can be generated:

| Days  | Susceptible                 | Infected              | Recovered/Removed |
|-------|-----------------------------|-----------------------|-------------------|
| t = 0 | $(1 - 1.76 \times 10^{-7})$ | $1.76 \times 10^{-7}$ | 0                 |

<sup>[6]</sup> Mahmud, A. H. (2021, September 17). *Singapore's COVID-19 reproduction number is above 1. What does this mean and should we be concerned?* CNA. Retrieved January 2, 2022, from <https://www.channelnewsasia.com/singapore/covid-19-r-number-reproduction-new-cases-2181166>

|       |                                                    |                                                                            |                                  |
|-------|----------------------------------------------------|----------------------------------------------------------------------------|----------------------------------|
| t = 1 | $S_0 + 0.107(S_0 \times I_0 \times 1)$             | $I_0 - (0.107 \times S_0 \times I_0) + (0.071 \times I_0)$                 | $R_0 + 0.071 \times I_0$         |
| t = 2 | $S_1 + 0.107(S_1 \times I_1 \times 1)$             | $I_1 - (0.107 \times S_0 \times I_1) + (0.071 \times I_1)$                 | $R_1 + 0.071 \times I_1$         |
| ...   | ...                                                | ...                                                                        | ...                              |
| t = n | $S_{n-1} + 0.107(S_{n-1} \times I_{n-1} \times 1)$ | $I_{n-1} - (0.107 \times S_{n-1} \times I_{n-1}) + (0.071 \times I_{n-1})$ | $R_{n-1} + 0.071 \times I_{n-1}$ |

By using Microsoft excel software, I applied the values from Table 1.1 (from t = 0 to t = 536) to plot the compartmental trends part of the SIR Model, as shown in *Figure 3.1* under the following subpart.<sup>[7]</sup>

### 5.3 Mathematical Formula Approximation for Singapore's SIR Model<sup>[8]</sup>

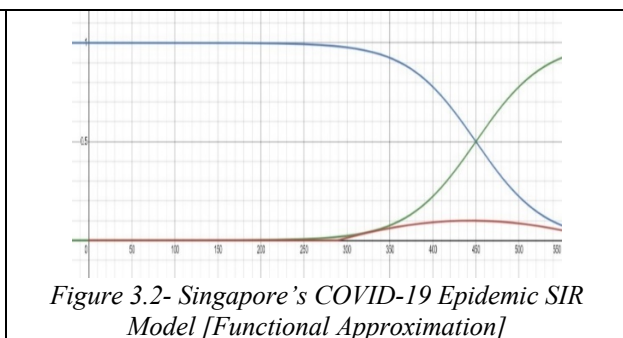
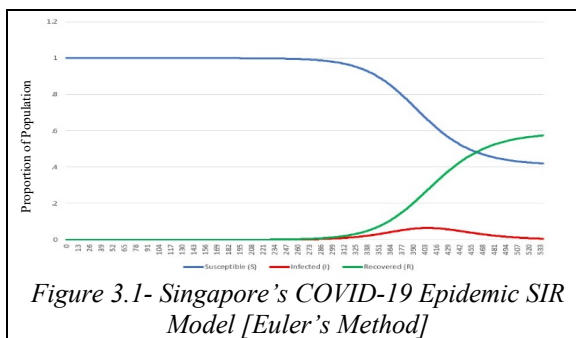
As previously stated in subsection 4.3 of this exploration, the trends can be modelled using Sigmoidal, Negative Sigmoidal, and Exponential Functions. Thus, the SIR Model for Singapore's COVID-19 Spread can also be mathematically approximated:

$$S(t) = -\frac{1}{1 + e^{-\frac{(x-450)}{40}}} + 1$$

$$I(t) = e^{-\left(\frac{x+31.318}{475}-1\right)^2} - 0.9$$

$$R(t) = \frac{1}{1 + e^{-\frac{x-450}{40}}}$$

Plotting these three functions along with the trend derived from Euler's Method, we get two comparable figures:



<sup>[7]</sup> Refer to appendix subsection 11.5 for raw values generated by Euler's method

<sup>[8]</sup> Note- Formula approximation has been included to mathematically describe SIR trends, but will not be used in the rest of the analysis for comparison to Singapore's Actual Trends



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## 6. SIR Model Analysis with respect to Singapore COVID-19 Outbreak

The predicted versus actual trends for the three compartments will be compared from the 23<sup>rd</sup> of January 2020 to the 11<sup>th</sup> of July 2021, during which time the epidemic engulfed Singapore.

To obtain the actual trend, I used the python's *matplotlib* library to plot the individual trends, based off a historical dataset of COVID-19 data in Singapore.<sup>[9][10]</sup> I then compared the individual trends obtained from this dataset to the predicted trends generated using Euler's Method.

### 6.1 Threshold Value Comparison

#### 6.1.1 Maximum Infection Density

Using the maximum infection density ( $I_{max}$ ) expression and  $R_0 = 1.5$ ,

$$\frac{I_{max}}{N} = 1 - \frac{1}{1.5}(\ln(1.5) + 1)$$
$$\frac{I_{max}}{N} = \mathbf{0.063}$$

Singapore Case Study Value for ( $\frac{I_{max}}{N}$ ) = **0.0037**  
( $0.0037 \neq 0.063$ \*)

#### 6.1.2 $S_\infty$ Value

Considering a subpart of the expression for  $S_\infty$ ,

$$S_\infty \geq S_0 e^{-R_0}$$

Substituting  $S_0 = 0.9999998241$  &  $R_0 = 1.5$ : ( $S_0$  value from Euler's Method)

$$\therefore S_\infty \geq 0.9999998241 \times e^{-1.5}$$

$$\mathbf{S_\infty \geq 0.22313}$$

Susceptible values approximately plateau once  $t = 536$ . Thus, the value for  $S(536)$  can be extrapolated to  $S_\infty$ .

$$\therefore \lim_{t \rightarrow 536} (S) = S_{536}; \approx S_\infty = \mathbf{0.98897} \text{ (Value obtained from COVID-19 dataset)}$$

( $0.22313 \neq 0.98897$ )

---

<sup>[9]</sup> Refer to appendix subsection 11.2 through 11.4 for code used

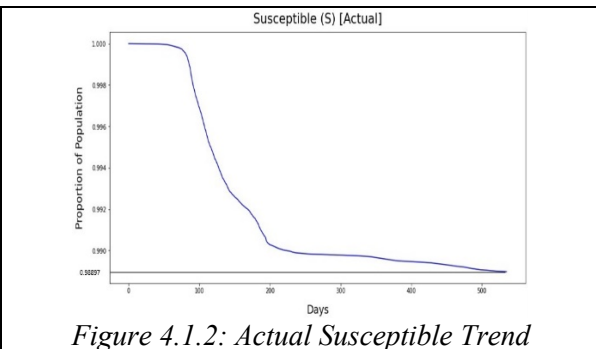
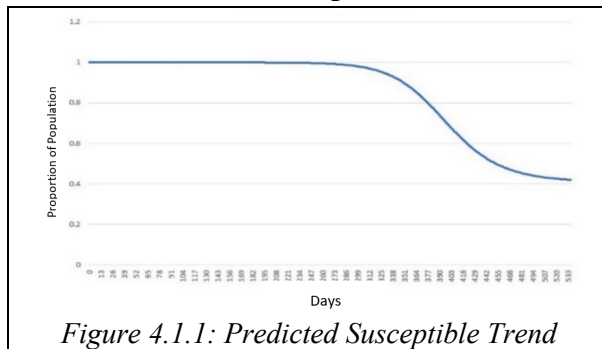
<sup>[10]</sup> Refer to citations for dataset used

The two sets of largely differing threshold values indicate that the predicted epidemic model differs largely from the actual case study values and trends. Aside from these threshold values, the trends themselves were in disaccord with the predicted model trends- the susceptible trend was predicted to plateau at around 0.420, however in Singapore it only reached as low as 0.989 and sharply declined early during the 536-day period. The infections were predicted to spike much later during the epidemic and reach a much higher  $I_{max}$  value as well as discussed in the previous subpart. Finally, the predicted *Recovered/Removed* compartment at  $t = 536$  was more than 50 times off  $R(536)$  in reality, at 0.011 compared to the predicted 0.574. These value disparities will be discussed in *evaluation* subsection 7.1.

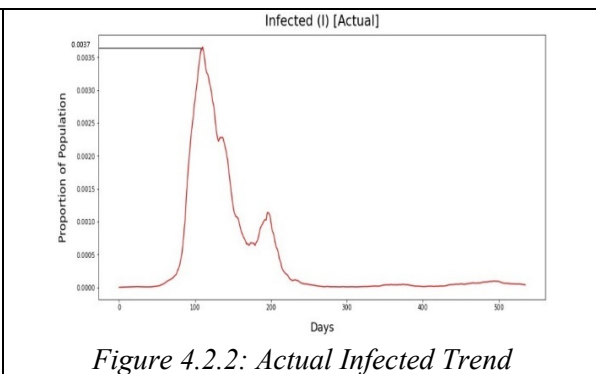
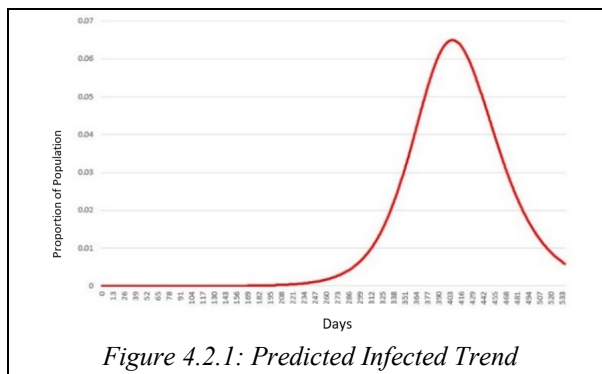
Numerically, the two models have been compared, but their shapes also need to be analyzed with respect to their function approximations and similarity.

## 6.2 Visual Trend Comparison

### 6.2.1 Susceptible Trend



### 6.2.2 Infected Trend



### 6.2.3 Recovered/Removed Trend

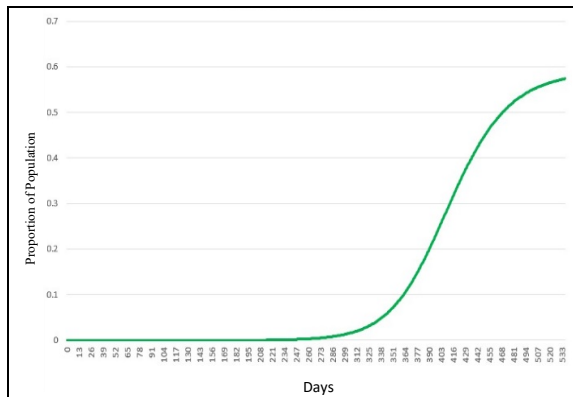


Figure 4.3.1: Recovered Predicted Trend

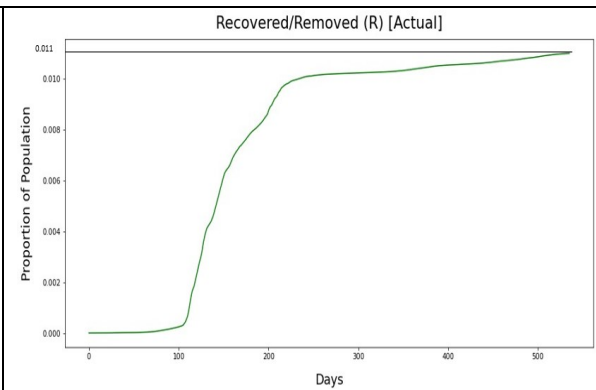


Figure 4.3.2: Recovered Actual Trend

Visually comparing the three pairs of trends, they are quite similar in nature and fall in accord with the mathematical formula approximations of the trends as discussed in subsection 4.3 & 5.3 of this exploration. Singapore's susceptible trend was indeed a negative sigmoidal function, the infected trend an  $e^{-x^2}$  line, and the recovered trend a sigmoidal function. However, the gradient of increase/decrease in the values was much larger in the predicted models than the actual numbers that transpired in Singapore, a fact which will be discussed in the next segment.

## 7. Evaluation

### 7.1 Analyzing Disparities with respect to Explorational Limitations

The previously stated disparities were likely caused by the limitations and unrealistic assumptions underlying SIR epidemic models. Firstly, Singapore's globally praised handling of the virus stopped population movement during the epidemic period, reducing the contacts between susceptible (S) and infected (I) individuals. Not only does this drastically reducing the value of  $\lambda$  calculated using compartmental epidemiology, but it disproves the assumption of a constant rate of contact. As a result, the original spike was quick to fall as Singapore almost completely stopped the spread of the virus in the country. Populations are also never constant throughout an epidemic period, the net birth rate as well as net migration would either add or remove people

from the population and thus the compartmental categories. Given that Euler's method is based on the value changes between each period ( $= \Delta x$ ), a change in one would affect all subsequent value and consequently generate a completely different model.

## 8. Compartmental Epidemiology Future Extensions

To generate more and more accurate models, several additional real-world factors need to be considered. For SIR modelling, this means the inclusion of additional compartmental categories to overcome initial unrealistic assumptions.

### 8.1 SIRD Epidemic Model

The SIRD Epidemic Model is an amendment to the SIR model which splits the recovered/removed category into *recoveries* and *deaths*, each having their own trendline. Thus, a new set of equations are obtained:

Recoveries/Removals  $\propto$  Infections  $\Rightarrow$  Recoveries  $\propto$  Infections & Deaths  $\propto$  Infections

$$D = m(I), \text{ for some constant } m \in R$$

$\therefore D = \mu(I)$ , where  $\mu$  is the fatality rate of the particular disease

$$D_{t+1} = D_t + \mu(I)$$

$$\therefore \frac{dD}{dt} = \mu(I)$$

Representing the 'death' compartment differential equation in Euler's form:

$$D_{t+1} = D_t + 1(\mu(I))$$

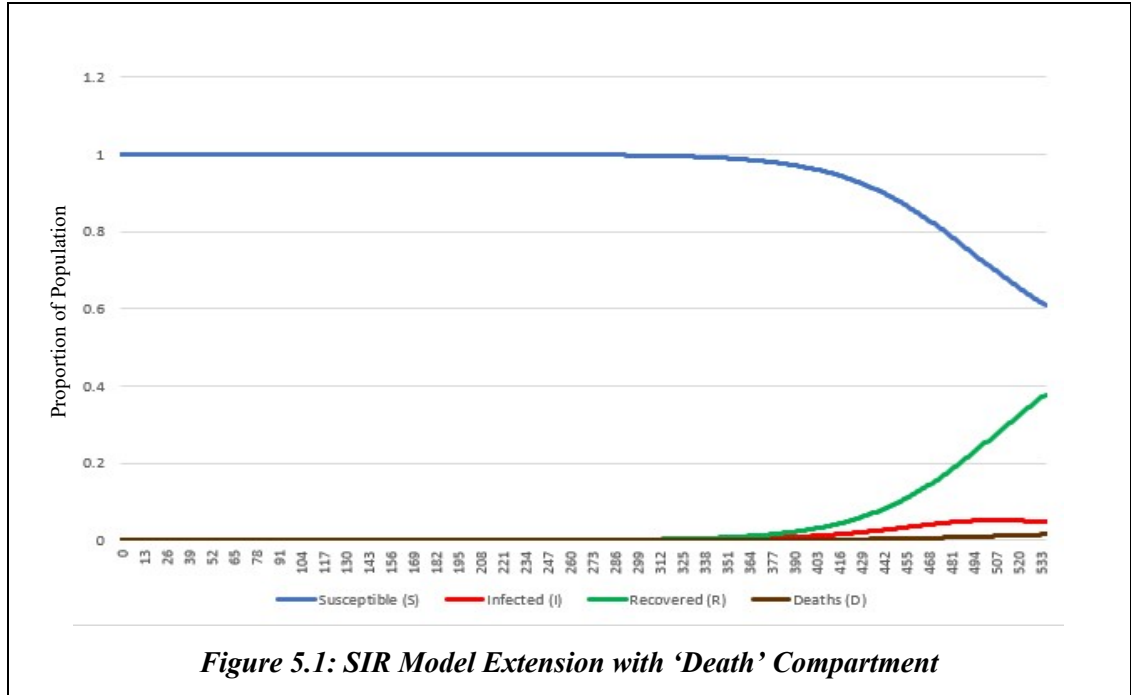
Given the assumption that  $S + I + R + D = N$ , the differential equations become:

$$\frac{dS}{dt} = -\lambda(SI), \frac{dI}{dt} = \lambda(SI) - \beta(I) - \mu(I), \frac{dR}{dt} = \beta(I), \text{ \& } \frac{dD}{dt} = \mu(I)$$

Applying the SIRD model to the chosen case study,  $\mu = 0.0028$ , given that COVID-19's mortality rate is around 0.28%<sup>[11]</sup>

$\beta$  no longer needs to be averaged for recoveries as well as deaths,  $\therefore \beta = \frac{1}{14} = 0.071$

Therefore, the SIRD model for COVID-19 spread in Singapore becomes:



## 8.2 SIRV Epidemic Model

Another variation to the base model is the addition of a category for *vaccinated* individuals, thus forming an SIRV model. The people in the vaccinated category move out of the susceptible category and do not risk entering the infected or recovered/removed category. In the same manner as the death category from the SIRD model extension,

$$V = n(S), \text{ for some constant } n \in R$$

$\therefore V = v(S)$ , where  $v$  is the *rate of vaccination of the susceptible population*

$$V_{t+1} = V_t + v(S)$$

<sup>[11]</sup> *Coronavirus Death Rate (COVID-19) - Worldometer.* (n.d.). Coronavirus (COVID-19) Mortality Rate. Retrieved February 22, 2022, from <https://www.worldometers.info/coronavirus/coronavirus-death-rate/>

$$\therefore \frac{dV}{dt} = \nu S$$

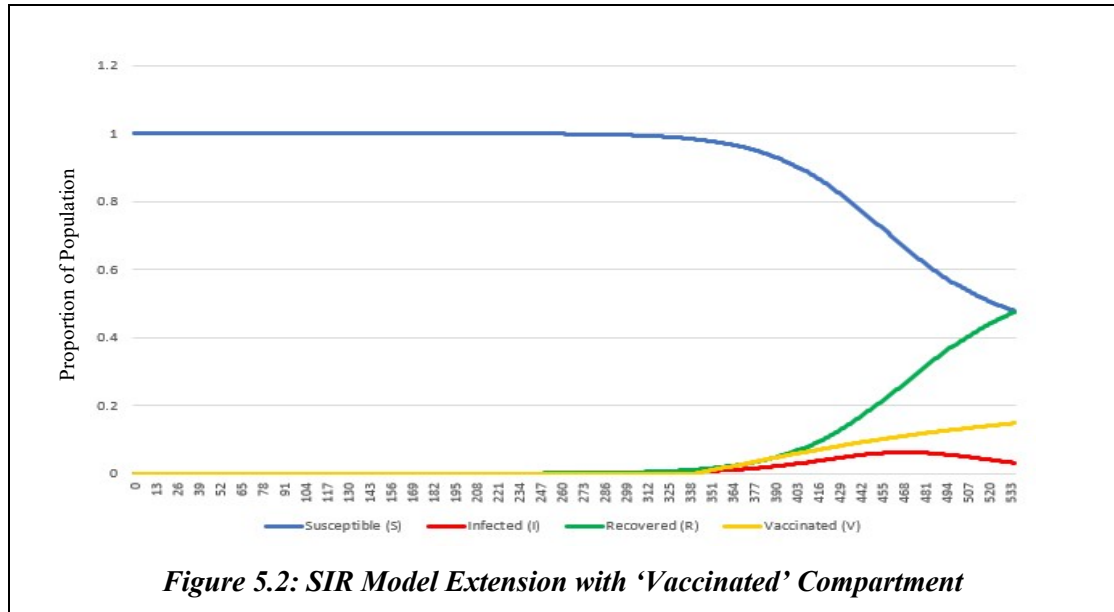
Representing the ‘Vaccinated’ compartment differential equation in Euler’s form:

$$V_{t+1} = V_t + 1 (\nu(S))$$

Given the assumption that  $S + I + R + D = N$ , the differential equations become:

$$\frac{dS}{dt} = -\lambda(SI) - \nu(S), \frac{dI}{dt} = \lambda(SI) - \beta(I) - \mu(I), \frac{dR}{dt} = \beta(I), \& \frac{dV}{dt} = \nu(S)$$

In the COVID-19 epidemic in Singapore, the first vaccination was given on the 30<sup>th</sup> of December 2020, 342 days after the start of the epidemic. Thus, plotting the SIRV trend using Euler’s method, the  $V$  category remains at 0 from  $t = 0$  to  $t = 342$ <sup>[12]</sup>.



### 8.3 SIR Epidemic Model with Vital Dynamics

Vital dynamics consider the birth rates ( $\Lambda$ ) and death rates ( $\Phi$ ) during the time period where the communicable disease is spreading, overcoming the unrealistic assumption of a standard population throughout the epidemic period ( $S + I + R \neq 1$ ). The inclusion of  $\Phi$  &  $\Lambda$  affects the spread of the population in all three compartments<sup>[13]</sup>.

<sup>[12]</sup> Due to lack of availability of data, the vaccination rate was fixed at  $\nu = 0.001$  for modelling purposes

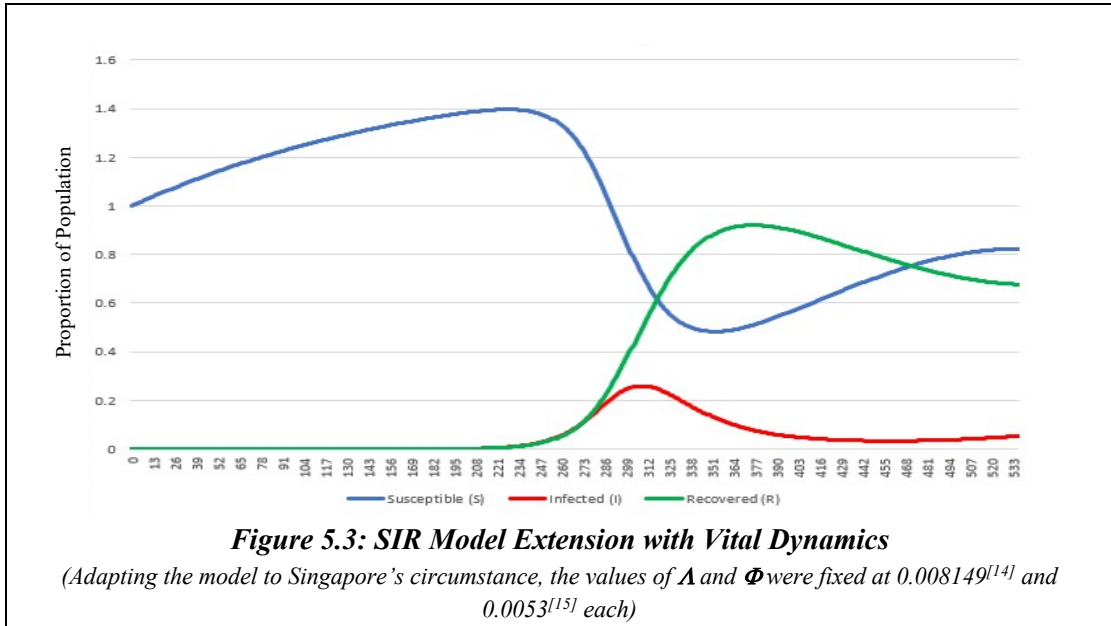
<sup>[13]</sup> Note- A new  $R_0$  value can be derived in terms of birth and death rate:  $R_0 = \frac{\lambda}{\Phi + \Lambda}$

$$\frac{dS}{dt} = \Lambda - \Phi(S) - \lambda(SI)$$

$$\frac{dR}{dt} = \beta(I) - \Phi(R)$$

$$\frac{dI}{dt} = \lambda(SI) - \beta(I) - \Phi(I)$$

[Negates first assumption,  $\therefore \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} \neq 0$ ]



#### 8.4 Hybrid SIRDV model with vital dynamics

Adding all the previously derived expressions into one, we get a set of expressions in terms of the time-dependent variables  $S(t)$ ,  $I(t)$ ,  $R(t)$ ,  $D(t)$ , and  $V(t)$ , and the constants  $\lambda$ ,  $\beta$ ,  $\mu$ ,  $\Lambda$ , and  $\Phi$ .

$$\frac{dS}{dt} = \Lambda - \Phi(S) - \lambda(SI) - v(S)$$

$$\frac{dI}{dt} = \lambda(SI) - \Phi(I) - \beta(I) - \mu(I)$$

$$\frac{dR}{dt} = \beta(I) - \Phi(R)$$

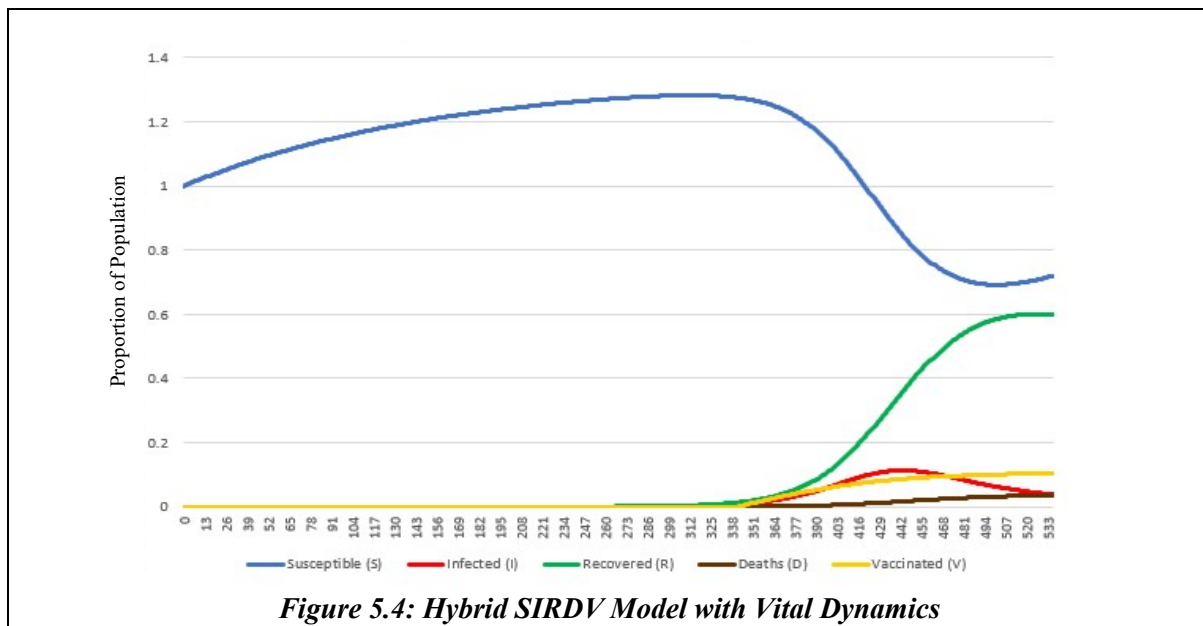
<sup>[14]</sup> *Coronavirus Death Rate (COVID-19) - Worldometer.* (2021, May 14). Coronavirus (COVID-19) Mortality Rate. Retrieved February 23, 2021, from <https://www.worldometers.info/coronavirus/coronavirus-death-rate/>

<sup>[15]</sup> *Auto, H.* (2021, November 6). *Singapore's overall death rate did not increase despite Covid-19.* The Straits Times. Retrieved February 23, 2021, from <https://www.straitstimes.com/singapore/singapores-overall-death-rate-did-not-increase-despite-covid-19>

$$\frac{dD}{dt} = \mu(I)$$

$$\frac{dV}{dt} = v(S) + \Phi(V)$$

Plotting a generic trend to see the hybrid curve, we get an epidemic model that looks like this:



## 9. Conclusion

In conclusion, I have been able to use this exploration to investigate the efficacy of SIR epidemic models and the far-reaching applications of epidemic theory. I was able to analytically quantify significant features of an epidemic within the model limitations, which can be used to predict epidemic impacts before they transpire, vital in fighting against them and limiting long-lasting negative impacts. Despite the fact that the generated model fell out of accord with Singapore's 2020-21 COVID-19 epidemic, I was able to analyze these shortcomings and briefly discussed ameliorations that could yield models of a greater accuracy in the future, such as vital dynamics and death/vaccine compartments. Working with Ordinary Differential Equations (ODEs) and applying Euler's method in several complex instances have strengthened my understanding of the topics, which have helped me better understand the questions I am exposed to as a part of the IB Mathematics Analysis and Approaches course.



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---

## 11. Appendix

### 11.1 Python Code Used in Exploration

#### 11.1.1 Graphing the generic trend of an SIR epidemic model<sup>[16]</sup>

##### 11.1.1.1 Importing the libraries

```
import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt
```

##### 11.1.1.2 Calculating the values of S, I, and R

```
N = 1000 #Total population, N
# Initial number of infected and recovered individuals,
I0 and R0.
I0, R0 = 1, 0
# Everyone else, S0, is susceptible to infection
initially.
S0 = N - I0 - R0
# Contact rate [beta] and mean recovery rate [gamma] (in
1/days).
beta, gamma = 0.2, 1/10
# A grid of time points (in days)
t = np.linspace(0, 160, 160)
# The SIR model differential equations.
def deriv(y, t, N, beta, gamma):
    S, I, R = y
    dSdt = -beta * S * I / N
    dIdt = beta * S * I / N - gamma * I
    dRdt = gamma * I
    return dSdt, dIdt, dRdt
# Initial conditions vector
y0 = S0, I0, R0
# Integrate the SIR equations over the time grid, t.
ret = odeint(deriv, y0, t, args=(N, beta, gamma))
S, I, R = ret.T
```

---

<sup>[16]</sup> Code adapted from: <https://scipython.com/book/chapter-8-scipy/additional-examples/the-sir-epidemic-model/>

---

### 11.1.1.3 Plotting the data using matplotlib

```
fig = plt.figure(facecolor='w', figsize = (15,7))
ax = fig.add_subplot(111, facecolor='#dddddd',
axisbelow=True)
ax.plot(t, S/1000, 'b', alpha=0.5, lw=2,
label='Susceptible')
ax.plot(t, I/1000, 'r', alpha=0.5, lw=2,
label='Infected')
ax.plot(t, R/1000, 'g', alpha=0.5, lw=2,
label='Recovered')
ax.set_xlabel('Time (days)', fontsize = 16)
ax.set_ylabel('Number of People', fontsize = 16)
ax.set_ylim(0,1.2)
ax.yaxis.set_tick_params(length=0)
ax.xaxis.set_tick_params(length=0)
ax.grid(b=True, which='major', c='w', lw=2, ls='-')
legend = ax.legend()
legend.get_frame().set_alpha(0.5)
for spine in ('top', 'right', 'bottom', 'left'):
    ax.spines[spine].set_visible(False)
plt.title("SIR Epidemic Model", fontsize = 22)
plt.show()
```

## 11.1.2 Opening the CSV file with daily case counts and deaths for the 2020 COVID-19 outbreak in Singapore

### 11.1.2.1 Importing the necessary libraries to read the data

```
import pandas as pd
import pylab as pl
import csv
```

### 11.1.2.2 Importing the raw data

```
SgData =
pd.read_csv(r'C:\Users\rohan\OneDrive\Rohan\IB\Subjects\Maths (MAA) (HL)\Maths IA\singapore_covid_data.csv')
SgData.columns = EbData.columns.str.replace(' ','_')
SgData.head(536)
```

---

### 11.1.3 Creating the data columns for the different SIR model components

#### 11.1.3.1 Creating a data column for the susceptible people

```
S = []  
# Singapore population in 2020: 5.686 million  
for i in range (536):  
    Val = (5686000 -  
int(SgData.loc[i, "Cumulative_Confirmed"])) / 5686000  
    S.append(Val)  
for i in range (536, 999):  
    S.append("0")  
SgData['Susceptible'] = S  
  
SgData.head(536)
```

#### 11.1.3.2 Creating a data column for recovered/remove people<sup>[17]</sup>

```
R = []  
for i in range (536):  
    Val = (int(SgData.loc[i, "Cumulative_Deaths"]) +  
int(SgData.loc[i, "Cumulative_Discharged"])) / 5686000  
    R.append(Val)  
for i in range(536, 999):  
    R.append("0")  
SgData['Recovered_Removed'] = R  
  
SgData.head(536)
```

#### 11.1.3.3 Creating a data column infected people

```
I = []  
for i in range (536):  
    Val = 1 - (SgData.loc[i, "Recovered_Removed"]) -  
(SgData.loc[i, "Susceptible"])  
    I.append(Val)  
for i in range (536, 999):  
    I.append("0")  
SgData['Infected'] = I
```

---

<sup>[17]</sup> *Susceptible Category = (Population Count in 2014) – (Daily Cases)*

---

```
SgData.head(536)
```

### 11.1.4 Plotting individual graphs for each trend

#### 11.1.4.1 Formatting the data columns to be plotted

```
Sus = SgData.Susceptible
Inf = SgData.Infected
Rem_Rec = SgData.Recovered_Removed
Dates = []
S = []
I = []
R = []
for i in range(536):
    Dates.append(i)
    S.append(Sus[i])
    I.append(Inf[i])
    R.append(Rem_Rec[i])
```

#### 11.1.4.2 Susceptible Trend

```
SusPlot = pd.DataFrame({'Date:': Dates, 'No. of Sus': S})
SusPlot.set_index('Date:')['No. of
Sus'].plot(figsize=(15, 7), linewidth=1.5, color='blue')

plt.xlabel("Days", labelpad=15, fontsize = 18)
plt.ylabel("Proportion of Population", labelpad=15,
fontsize = 18)
plt.title("Susceptible (S) [Actual]", y=1.02,
fontsize=22)
```

#### 11.1.4.3 Infected Trend

```
InfPlot = pd.DataFrame({'Date:': Dates, 'No. of Inf': I})
InfPlot.set_index('Date:')['No. of
Inf'].plot(figsize=(15, 7), linewidth=1.5, color='red')

plt.xlabel("Days", labelpad=15, fontsize = 18)
plt.ylabel("Proportion of Population", labelpad=15,
fontsize = 18)
plt.title("Infected (I) [Actual]", y=1.02, fontsize=22)
```

#### 11.1.4.4 Recovered/Removed Trend

```
RemPlot = pd.DataFrame({'Date': Dates, 'No. of Rem': R})
RemPlot.set_index('Date')['No. of
Rem'].plot(figsize=(15, 7), linewidth=1.5, color='green')

plt.xlabel("Days", labelpad=15, fontsize = 18)
plt.ylabel("Proportion of Population", labelpad=15,
fontsize = 18)
plt.title("Recovered/Removed (R) [Actual]", y=1.02,
fontsize=22)
```

### 11.2 Raw data generated by Euler's Method

#### 11.2.1 Singapore's COVID-19 Epidemic SIR Model

| Days (t) | Susceptible (S) | Infected (I) | Recovered (R) |
|----------|-----------------|--------------|---------------|
| 0        | 0.999999824     | 1.759E-07    | 0             |
| 1        | 0.999999805     | 1.82232E-07  | 1.24889E-08   |
| 2        | 0.999999786     | 1.88793E-07  | 2.54274E-08   |
| 3        | 0.999999766     | 1.95589E-07  | 3.88317E-08   |
| 4        | 0.999999745     | 2.02631E-07  | 5.27185E-08   |
| 5        | 0.999999723     | 2.09925E-07  | 6.71053E-08   |
| 6        | 0.999999701     | 2.17482E-07  | 8.201E-08     |
| 7        | 0.999999677     | 2.25312E-07  | 9.74512E-08   |
| 8        | 0.999999653     | 2.33423E-07  | 1.13448E-07   |
| 9        | 0.999999628     | 2.41826E-07  | 1.30021E-07   |
| 10       | 0.999999602     | 2.50532E-07  | 1.47191E-07   |

:

:

:

:

|     |             |             |             |
|-----|-------------|-------------|-------------|
| 527 | 0.422946383 | 0.007424481 | 0.569629136 |
| 528 | 0.422610386 | 0.00723334  | 0.570156274 |
| 529 | 0.4222833   | 0.007046859 | 0.570669841 |
| 530 | 0.421964892 | 0.00686494  | 0.571170168 |
| 531 | 0.421654938 | 0.006687483 | 0.571657579 |
| 532 | 0.421353219 | 0.006514391 | 0.57213239  |
| 533 | 0.421059519 | 0.00634557  | 0.572594912 |

|     |             |             |             |
|-----|-------------|-------------|-------------|
| 534 | 0.420773629 | 0.006180923 | 0.573045447 |
| 535 | 0.420495347 | 0.00602036  | 0.573484293 |
| 536 | 0.420224473 | 0.005863789 | 0.573911738 |

### 11.2.2 SIRD Epidemic Model

| Days (t) | Susceptible (S) | Infected (I) | Recovered (R) | Deaths (D)  |
|----------|-----------------|--------------|---------------|-------------|
| 0        | 0.999999824     | 1.759E-07    | 0             | 0           |
| 1        | 0.999999805     | 1.8174E-07   | 1.24889E-08   | 4.9252E-10  |
| 2        | 0.999999786     | 1.87774E-07  | 2.53924E-08   | 1.00139E-09 |
| 3        | 0.999999766     | 1.94008E-07  | 3.87244E-08   | 1.52716E-09 |
| 4        | 0.999999745     | 2.00449E-07  | 5.24989E-08   | 2.07038E-09 |
| 5        | 0.999999724     | 2.07104E-07  | 6.67308E-08   | 2.63164E-09 |
| 6        | 0.999999701     | 2.13979E-07  | 8.14351E-08   | 3.21153E-09 |
| 7        | 0.999999678     | 2.21084E-07  | 9.66277E-08   | 3.81067E-09 |
| 8        | 0.999999655     | 2.28424E-07  | 1.12325E-07   | 4.4297E-09  |
| 9        | 0.99999963      | 2.36007E-07  | 1.28543E-07   | 5.06929E-09 |
| 10       | 0.999999605     | 2.43843E-07  | 1.45299E-07   | 5.73011E-09 |
| ⋮        | ⋮               | ⋮            | ⋮             | ⋮           |

|     |             |             |             |             |
|-----|-------------|-------------|-------------|-------------|
| 527 | 0.479099892 | 0.01442814  | 0.487256229 | 0.019215739 |
| 528 | 0.478360252 | 0.014102983 | 0.488280627 | 0.019256137 |
| 529 | 0.477638398 | 0.013784038 | 0.489281939 | 0.019295626 |
| 530 | 0.476933932 | 0.013471241 | 0.490260606 | 0.019334221 |
| 531 | 0.476246469 | 0.013164527 | 0.491217064 | 0.019371941 |
| 532 | 0.475575626 | 0.012863827 | 0.492151745 | 0.019408801 |
| 533 | 0.47492103  | 0.012569073 | 0.493065077 | 0.01944482  |
| 534 | 0.474282313 | 0.012280193 | 0.493957481 | 0.019480013 |
| 535 | 0.473659115 | 0.011997112 | 0.494829375 | 0.019514398 |
| 536 | 0.473051083 | 0.011719757 | 0.49568117  | 0.01954799  |



### 11.2.3 SIRV Epidemic Model

| Days (t) | Susceptible (S) | Infected (I) | Recovered (R) | Vaccinated (V) |
|----------|-----------------|--------------|---------------|----------------|
| 0        | 0.999999824     | 1.759E-07    | 0             | 0              |
| 1        | 0.999999805     | 1.82232E-07  | 1.24889E-08   | 0              |
| 2        | 0.999999786     | 1.88793E-07  | 2.54274E-08   | 0              |
| 3        | 0.999999766     | 1.95589E-07  | 3.88317E-08   | 0              |
| 4        | 0.999999745     | 2.02631E-07  | 5.27185E-08   | 0              |
| 5        | 0.999999723     | 2.09925E-07  | 6.71053E-08   | 0              |
| 6        | 0.999999701     | 2.17482E-07  | 8.201E-08     | 0              |
| 7        | 0.999999677     | 2.25312E-07  | 9.74512E-08   | 0              |
| 8        | 0.999999653     | 2.33423E-07  | 1.13448E-07   | 0              |
| 9        | 0.999999628     | 2.41826E-07  | 1.30021E-07   | 0              |
| 10       | 0.999999602     | 2.50532E-07  | 1.47191E-07   | 0              |

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|-----|-------------|-------------|-------------|-------------|
| 527 | 0.415605082 | 0.007055324 | 0.565507999 | 0.111402382 |
| 528 | 0.415179931 | 0.006868144 | 0.566008927 | 0.111817987 |
| 529 | 0.414763001 | 0.006685618 | 0.566496565 | 0.112233167 |
| 530 | 0.414354062 | 0.006507645 | 0.566971244 | 0.11264793  |
| 531 | 0.413952892 | 0.006334124 | 0.567433287 | 0.113062284 |
| 532 | 0.413559273 | 0.006164958 | 0.56788301  | 0.113476237 |
| 533 | 0.413172992 | 0.006000051 | 0.568320722 | 0.113889796 |
| 534 | 0.412793843 | 0.005839307 | 0.568746725 | 0.114302969 |
| 535 | 0.412421624 | 0.005682632 | 0.569161316 | 0.114715763 |
| 536 | 0.412056139 | 0.005529934 | 0.569564783 | 0.115128185 |

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#### 11.2.4 SIR Epidemic Model with Vital Dynamics

| Days (t) | Susceptible (S) | Infected (I) | Recovered (R) |
|----------|-----------------|--------------|---------------|
| 0        | 0.999999824     | 1.759E-07    | 0             |
| 1        | 1.003118806     | 1.813E-07    | 1.24889E-08   |
| 2        | 1.006221257     | 1.86927E-07  | 2.5295E-08    |
| 3        | 1.009307264     | 1.9279E-07   | 3.84327E-08   |
| 4        | 1.012376915     | 1.989E-07    | 5.19171E-08   |
| 5        | 1.015430296     | 2.0527E-07   | 6.57639E-08   |
| 6        | 1.018467493     | 2.11911E-07  | 7.99895E-08   |
| 7        | 1.021488592     | 2.18835E-07  | 9.46112E-08   |
| 8        | 1.024493679     | 2.26056E-07  | 1.09647E-07   |
| 9        | 1.027482837     | 2.33589E-07  | 1.25116E-07   |
| 10       | 1.030456153     | 2.41447E-07  | 1.41038E-07   |

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|-----|-------------|-------------|-------------|
| 527 | 0.422946383 | 0.007424481 | 0.569629136 |
| 528 | 0.422610386 | 0.00723334  | 0.570156274 |
| 529 | 0.4222833   | 0.007046859 | 0.570669841 |
| 530 | 0.421964892 | 0.00686494  | 0.571170168 |
| 531 | 0.421654938 | 0.006687483 | 0.571657579 |
| 532 | 0.421353219 | 0.006514391 | 0.57213239  |
| 533 | 0.421059519 | 0.00634557  | 0.572594912 |
| 534 | 0.420773629 | 0.006180923 | 0.573045447 |
| 535 | 0.420495347 | 0.00602036  | 0.573484293 |
| 536 | 0.420224473 | 0.005863789 | 0.573911738 |

### 11.2.5 Hybrid SIRDV Epidemic Model with Vital Dynamics

| Days<br>(t) | Susceptible (S) | Infected<br>(I) | Recovered<br>(R) | Deaths<br>(D) | Vaccinated (V) |
|-------------|-----------------|-----------------|------------------|---------------|----------------|
| 0           | 0.999999824     | 1.759E-07       | 0                | 0             | 0              |
| 1           | 1.002118806     | 1.80808E-07     | 1.24889E-08      | 4.9252E-10    | 0              |
| 2           | 1.004224439     | 1.85893E-07     | 2.526E-08        | 9.98781E-10   | 0              |
| 3           | 1.006316805     | 1.91164E-07     | 3.83246E-08      | 1.51928E-09   | 0              |
| 4           | 1.008395988     | 1.96626E-07     | 5.16941E-08      | 2.05454E-09   | 0              |
| 5           | 1.010462072     | 2.02289E-07     | 6.53806E-08      | 2.60509E-09   | 0              |
| 6           | 1.012515139     | 2.08159E-07     | 7.93965E-08      | 3.1715E-09    | 0              |
| 7           | 1.014555271     | 2.14245E-07     | 9.3755E-08       | 3.75435E-09   | 0              |
| 8           | 1.01658255      | 2.20557E-07     | 1.0847E-07       | 4.35423E-09   | 0              |
| 9           | 1.018597056     | 2.27101E-07     | 1.23554E-07      | 4.97179E-09   | 0              |
| 10          | 1.02059887      | 2.33889E-07     | 1.39024E-07      | 5.60768E-09   | 0              |

|     |             |                 |                 |                 |             |
|-----|-------------|-----------------|-----------------|-----------------|-------------|
| 527 | 0.804071046 | 0.0298078<br>57 | 0.5520123<br>1  | 0.050274<br>558 | 0.080675926 |
| 528 | 0.804859862 | 0.0300145<br>93 | 0.5512030<br>02 | 0.050358<br>02  | 0.081052415 |
| 529 | 0.805623388 | 0.0302252<br>95 | 0.5504126<br>62 | 0.050442<br>061 | 0.081427697 |
| 530 | 0.806361489 | 0.0304399<br>46 | 0.5496414<br>71 | 0.050526<br>692 | 0.081801753 |

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|     |             |                 |                 |                 |             |
|-----|-------------|-----------------|-----------------|-----------------|-------------|
| 531 | 0.807074032 | 0.0306585<br>26 | 0.5488896<br>08 | 0.050611<br>924 | 0.082174566 |
| 532 | 0.80776089  | 0.0308810<br>12 | 0.5481572<br>48 | 0.050697<br>768 | 0.082546114 |
| 533 | 0.808421937 | 0.0311073<br>83 | 0.5474445<br>67 | 0.050784<br>235 | 0.082916381 |
| 534 | 0.809057055 | 0.0313376<br>13 | 0.5467517<br>35 | 0.050871<br>335 | 0.083285346 |
| 535 | 0.809666126 | 0.0315716<br>77 | 0.5460789<br>21 | 0.050959<br>081 | 0.083652991 |
| 536 | 0.81024904  | 0.0318095<br>47 | 0.5454262<br>92 | 0.051047<br>481 | 0.084019296 |